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An initiative of **PHP** Family
University of Information Technology & Sciences

Project Proposal

Software Engineering and System Analysis Lab

CSE 356

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Project Title

TourZ – Smart Online Tour & Travel Management System

Problem Statement

Traditional tour booking systems in Bangladesh rely heavily on manual communication, social media confirmations, and unstructured payment handling. This results in booking errors, poor user experience, lack of automated ticket generation, no real-time admin control, inefficient custom tour handling, and absence of 24/7 support.

TourZ aims to solve these issues through a centralized, web-based platform that automates booking, QR-based ticketing, PDF generation, chatbot support, and administrative management.

Objectives

- Develop a dynamic tour booking system using PHP and MySQL.
- Implement secure authentication and profile management.
- Generate QR-based booking verification and downloadable PDF tickets.
- Integrate a Python-based chatbot for 24/7 automated support.
- Enable users to request new tours and locations.
- Provide a complete admin panel for managing users, bookings, and tours.
- Maintain structured and normalized database design (`data.sql`).

Project Planning

Weeks 1–3:

Proposal preparation, presentation, and approval.

Requirement Analysis (Week 4):

User: Booking, payment, QR ticket, PDF download, chatbot support.

Admin: Tour management, user control, booking monitoring.

System Design (Weeks 4–10):

- Frontend: HTML, CSS, JavaScript
- Backend: PHP (core system), Python (chatbot service)
- Database: MySQL
- Integration: REST API (PHP ↔ Python)

- QR Module: phpqrcode
- Diagrams: ERD, DFD, UML, Gantt Chart

Development (Weeks 5–11):

Database & Authentication → Booking Module → QR & Payment → Admin Panel → PDF System → Chatbot Integration → Deployment

Testing (Weeks 12–13):

Unit, Integration, Security, and User Acceptance Testing

Features

User Features

- Registration & Login
- Tour Browsing & Booking
- Secure Payment Interface
- QR Code Ticket
- Downloadable PDF Ticket
- Custom Tour Request
- AI Chatbot Support

Admin Features

- Tour & User Management
- Booking Control
- Custom Request Approval
- Chatbot Query Monitoring
- Report Generation

Technology

Frontend: HTML5, CSS3, JavaScript

Backend: PHP, Python

Database: MySQL

Libraries: phpqrcode, FPDF/TCPDF

Framework (Chatbot API): Flask / FastAPI

Architecture: Client–Server, Modular PHP, REST API Integration

SWOT Analysis

Strengths: Automation, QR validation, chatbot support, modular design.

Weaknesses: Demo payment system, hosting dependency.

Opportunities: bKash/Nagad integration, mobile app, AI recommendation.

Threats: Security risks, market competition, server downtime.

Conclusion

TourZ digitizes traditional tour booking by integrating authentication, automated booking, QR verification, PDF ticketing, and Python-based chatbot support into a centralized system. The project demonstrates practical application of software engineering concepts including modular design, API integration, database management, and software testing. It has strong potential for commercial scalability with secure payment gateway and cloud deployment.

Reference

- PHP Documentation – <https://www.php.net>
- MySQL Documentation – <https://dev.mysql.com/doc>
- Python Documentation – <https://docs.python.org>
- Flask Documentation – <https://flask.palletsprojects.com>
- Pressman, *Software Engineering: A Practitioner's Approach*, 7th Edition